VF Asia Footwear Laboratory Testing Manual

Test Method for Safety Toecap – Impact Resistance	Test method # VF- B28 Page 1 of 2 Ref #: TM-R----
Date Issued: April 30, 2019 Revision #:	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To test resistance of safety footwear toecap to impact. The dart controlled by an electromagnet drops from a specified height and hits the toecap so that the damage to the toecap is measured.

REFERENCES:

1. ASTM F2413-18, Section5.2
2. CSA Z195-14, Section 5.1
3. NOM-113-STPS-2009, Section 6.2.2
4. ISO 20345:2011, Section 5.3.2.3

EQUIPMENT AND MATERIAL:

1. Toe Impact Test apparatus meeting the requirements of ASTM F2413-18, CSA Z195-14, NOM -113-STPS-2009, ISO 20345:2011
2. Finished unworn protective footwear
3. Digital caliper capable of measuring up to 0.01mm
4. Standard straight steel ruler
5. Cutter, such as sharp knife or scalpel
6. Plasticine, modeling clay
7. Aluminum foil, waxed paper or equivalent non-stick material
8. Marking pen

SAMPLE PREPARATION:

1. Randomly select three half-pair test specimens, including both left and right footwear.
2. Condition the protective footwear for at least 24 hours at standard laboratory conditions at a temperature of 23.0±2.0 ˚C and a relative humidity at 55 ±10 %.

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3. The sample is obtained by completely removing the toe portion of the footwear. This is done by cutting across the width of the footwear behind the back edge of the protective toe cap and cutting the upper material back to the edge of the protective toe cap. ***Note**: **The distance from the back edge of the toecap** - refer to standard as specified by ASTM F2413-18,* CSA Z195-14, *NOM -113-STPS-2009, ISO 20345:2011*

REPETITIONS: Test three (3) half - pair

TEST PROCEDURE:

Test methods – refer to applicable industry standards as specified in Table 1.

Table 1. VF Footwear Safety Shoe Requirements for Impact Resistance:

Test	Applicable Industry Standard, Including Requirements
Safety Toecap - Impact Resistance	• ASTM F2413 – 18, *Section 5.2* • CSA Z195-14, *Section 5.1* • NOM-113-STPS-2009, *Section 6.2.2* • ISO 20345:2011, *Section 5.3.2.3*